



Meta-analysis of the effects of age at first calving on production outcomes, calving difficulty, and reproduction in dairy heifers

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ABSTRACT

This study aims to determine the effects of feeding, management, and reproductive strategies used to reduce age at first calving (AFC) on production and reproductive performance through meta-analytic methods. A literature search using 3 search engines was conducted to identify prospective studies that evaluated AFC for data extraction. Inclusion criteria included specific dairy breeds, a treatment group with an AFC below 26 mo of age, and prospective enrollment of heifers. There were few prospective studies suitable for meta-analysis, with 16 studies containing 35 appropriate experiments. There was substantial heterogeneity, likely due to differences in study design and interventions used to induce an AFC difference. Outcomes were assessed using classical multilevel random effects meta-analytic models with standard mean difference, effect sizes, or risk ratios being evaluated where appropriate. Meta-regression was performed using the difference in AFC. There was a loss of production for first-lactation milk (−2.06 L/d of earlier calving), fat (−0.12 kg/d of earlier calving), and protein (−0.08 of kg/d earlier calving) yield for heifers calving at a younger age than their peers. There were no differences in second- and third-lactation milk yield, calving difficulty, heifer reproductive performance, survival to calving, and survival to the end of first lactation. Although the production in first lactation is decreased, this may be offset by fewer days to first calving without compromising the heifers' ability to calve, become pregnant for the first time, or survive to the end of first lactation. Given the lack of prospective studies, more randomized controlled experiments over multiple lactations, including health, production, and reproductive data, are needed to evaluate this topic further.

Key words: age at first calving, calving, dairy heifer, lactation, meta-analysis

INTRODUCTION

Lowering the age at first calving (AFC) in dairy heifers has long been viewed as a potential method to improve productivity in the dairy industry. Traditionally, calving was targeted for the spring flush in pasture-based systems, with the target AFC being 24 mo of age (Berry and Cromie, 2009). In the last 60 yr, housing of cattle and year-round calving systems have led to increased investigation into enhanced heifer feeding to increase growth rates, resulting in a decrease in the AFC (Swanson, 1961).

The objectives of decreased AFC are an earlier return on investment (Hutchison et al., 2017), improved longevity (Han et al., 2022), decreased GHG emissions (Dall-Orsoletta et al., 2019), decreased generation interval, improved reproductive performance (Lin et al., 1986), and increased lifetime milk production (Krpáľková et al., 2014). Potential risks include increased dystocia (Erb et al., 1985), fatty udder syndrome (Sejrsen et al., 2000), decreased first-lactation production (Lin et al., 1986), and increased stillbirths (Ettema and Santos, 2004; Hutchison et al., 2017).

The industry is trending toward a younger AFC internationally. Hutchison et al. (2017) reported a 4-fold increase in the percentage of Holstein heifers in the USA calving at 22 mo of age compared with 1997, with a similar increase in early calving in the Jersey breed. A previous meta-analysis investigated the effects of varying prepubertal growth rates on first-lactation production (Zanton and Heinrichs, 2005). Prepubertal growth rates of 836 g/d were optimal for future milk production in the first lactation, suggesting there is a limit on how early dairy heifers can calve and perform well. Studies have variable recommendations for the youngest AFC target, from 20.5 to 22 mo (Lin et al., 1988; Curran et al., 2013; Heinrichs et al., 2017; Hutchison et al., 2017; Eastham

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The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-26. Nonstandard abbreviations are available in the Notes.

et al., 2018) up to 26 mo of age (Froidmont et al., 2013; Turiello et al., 2020).

Retrospective analysis of AFC is the most reported way to examine this strategy. This approach can increase bias and confounding in the results. Without control over the breeding, growth management, or calving age, or all of these, it is difficult to interpret production and reproductive outcomes. For example, if heifers are bred by accident and calve early without sufficient growth, they are likely to have decreased production. Similarly, heifers that calve late due to poor health may also have abnormal production outcomes. These anomalies may confound and bias retrospective study results. Examination of the prospective literature through meta-analytic methods will provide further insight into AFC and minimize some potential bias likely in retrospective analysis.

We hypothesize that feeding, management, and breeding strategies that lead to a reduction in AFC will reduce the milk volume, fat, and protein production of the heifers in first lactation but have no ill effects on calving ease at first calving or reproductive performance. Later lactation milk volume, fat, and protein production will not be affected by interventions that reduce AFC.

MATERIALS AND METHODS

Search of the Published Literature

We used a systematic search of the literature consistent with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) methodology (Page et al., 2021). The literature search included 3 search engines, SCOPUS (<https://www.scopus.com/>), Google Scholar (<http://scholar.google.com/>), and the ISI Web of Science (<http://apps.webofknowledge.com>). An initial search was conducted between November 9 and 14, 2023, with an additional search occurring on November 8, 2024, for papers between 2023 and November 2024. For SCOPUS, the following search was used: Age at first calving OR Age of first calving OR Age at first mating OR Age at first breeding OR AFC AND Dairy. The search used for Web of Science and Google Scholar was “Age at first calving” OR “Age of first calving” OR “Age at first mating” OR “Age at first breeding” OR “AFC” AND “Prospective” AND “Dairy.” Google Scholar identified several thousand articles with this search strategy. These were sorted by relevance, and the assessment of titles was stopped after the evaluation of 6 consecutive pages of 10 titles without an appropriate study identified for abstract assessment (a total of 71 pages of results were assessed in 2023, and a total of 17 pages were assessed in 2024).

Relevant review articles and a meta-analysis were assessed for potential articles. These were identified

through the PRISMA search and references from the final articles. The references from included articles were also assessed for potential use.

Experiment Inclusion and Exclusion

The initial inclusion criteria were for experiments with titles of relevance to dairy heifer AFC or heifer feeding management or growth. Exclusion criteria were *Bos indicus*, beef cattle, retrospective studies, retrospective enrollment of heifers, no treatment AFC groups <26 mo of age, articles >40 yr old (from the initial 2023 search), not peer-reviewed, nonrandomized studies, and non-English language titles. The articles were also assessed for clear records of experimental numbers, measures of effect amenable to effect size analysis, and measures of variance. Any duplicate articles were removed. Responses of interest were milk production, milk fat and protein production, health outcomes, reproductive performance, calving difficulties, and growth rates.

Two authors reviewed the abstracts, followed by the full articles, for the above criteria. When a disagreement arose between the 2 reviewers, a third person assessed the abstract or article. Once the final articles were selected, their references were screened for further potential articles, which then underwent the same review process. Figure 1 depicts the PRISMA flow diagram (Page et al., 2021) from the data collection process.

The initial search of Web of Science, Google Scholar, and SCOPUS returned 1,539, 19,700 (710 assessed), and 823 articles, whereas the 2024 search returned 179, 3,280 (170 assessed), and 81 articles, respectively. The initial screening for the title identified 711 documents for abstract assessment. After abstract assessment, 128 articles were assessed in full. Articles were excluded for the following reasons: abstract only (3), no AFC groups calving below 26 mo of age (4), abstract in English but article in another language (18), off-topic (10), retrospective analysis or enrollment (68), review (3), study design was not randomized (5), and same study as another assessed (1). The final number of studies included was 16, consisting of 35 experiments (Table 1).

Data Extraction

Data were extracted and transferred to Excel spreadsheets (Microsoft 365 version, Microsoft Corp., Redmond, WA) and reviewed by 3 authors for accuracy before undertaking statistical analysis. Descriptive data extracted are summarized in Table 1. These data included number of animals in control and treatment groups, whether the experiment was randomized, if the experiment was pseudoreplicated, breed of cattle, housing system, intervention leading to AFC difference, age of in-

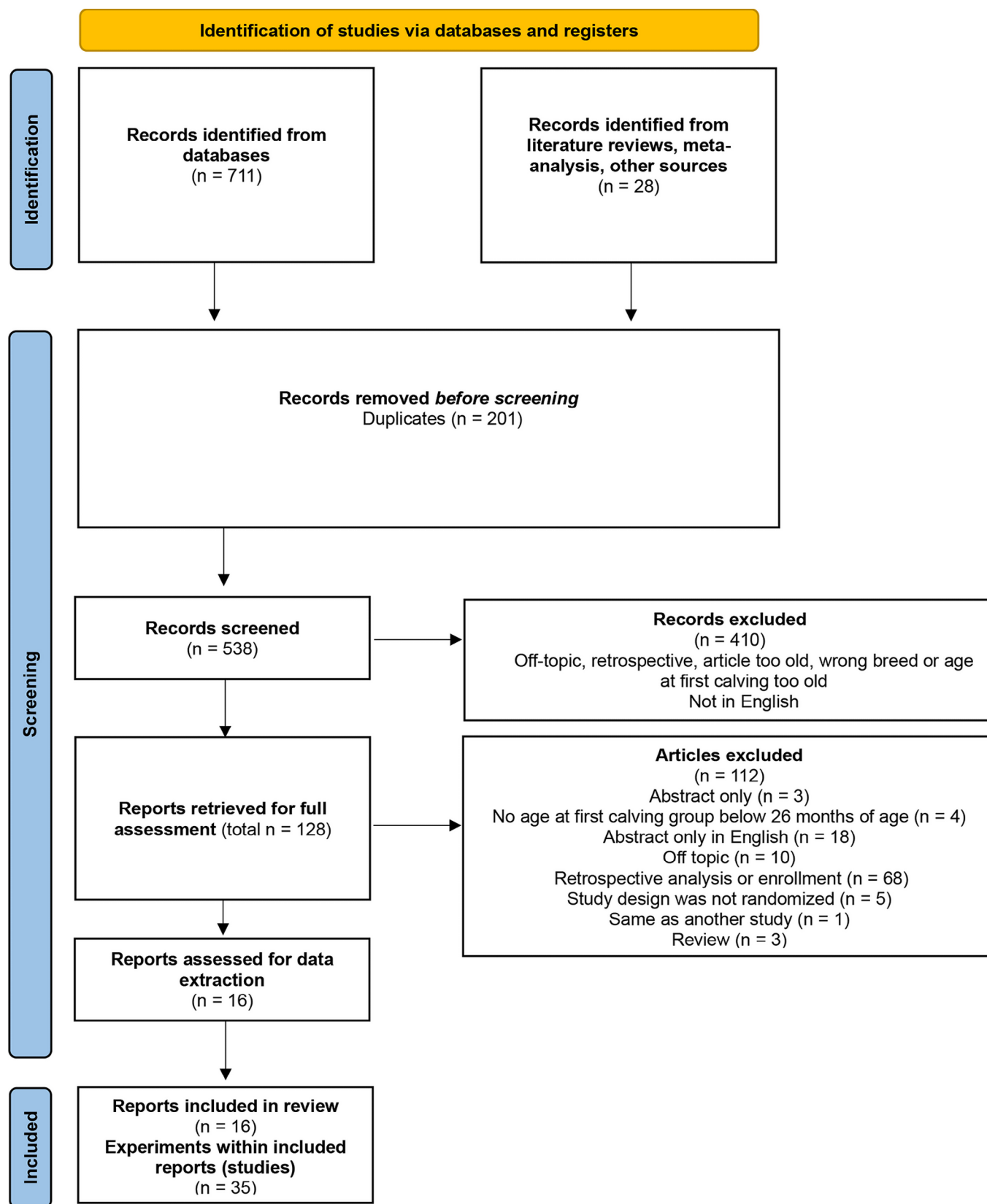


Figure 1. The PRISMA flow diagram (adapted from Page et al., 2021) of the initial systematic review and screening to final selection for meta-analysis. The n refers to the number of records identified for screening that include experimental articles, abstracts, books, review articles, and others.

Table 1. Summary of descriptive details of each experiment included in the meta-analysis on the intervention of age at first calving (AFC)¹

Experiment	No. of animals			Randomized	Pseudoreplicated	Breed	Housing system ²	Intervention	Intervention			AFC difference (d)
	C	T							Start age ³ (d)	Finish ³ (d)	AFC ^{4,5}	
Bar-Peled et al. (1997)	20	20	Y	Y	N	Holstein	Intensive	Conventional rearing vs. suckled calves	5	42	700, 669	31
Davis Rincker et al. (2011)	40	40	Y	Y	N	Holstein	Intensive	Conventional calf feeding vs. intensive calf feeding	2	42	715, 701	14
Dobos et al. (2001)-1	15	15	Y	Y	N	HF	Pasture	AFC/age at breeding and weight at calving	112	Calving	1,026, 771	255
Dobos et al. (2001)-2	15	15	Y	Y	N	HF	Pasture	AFC/age at breeding and weight at calving	112	Calving	1,026, 918	108
Dobos et al. (2001)-3	15	15	Y	Y	N	HF	Pasture	AFC/age at breeding and weight at calving	112	Calving	1,005, 744	261
Dobos et al. (2001)-4	15	15	Y	Y	N	HF	Pasture	AFC/age at breeding and weight at calving	112	Calving	1,005, 891	114
Dobos et al. (2001)-5	15	15	Y	Y	N	HF	Pasture	AFC/age at breeding and weight at calving	112	Calving	1,020, 744	276
Dobos et al. (2001)-6	15	15	Y	Y	N	HF	Pasture	AFC/age at breeding and weight at calving	112	Calving	1,020, 879	141
Dobos et al. (2004)-1	45	45	Y	Y	N	HF	Pasture	AFC/age at breeding and weight at calving	112	Calving	1,017, 753	264
Dobos et al. (2004)-2	45	45	Y	Y	N	HF	Pasture	AFC/age at breeding and weight at calving	112	Calving	1,017, 897	120
Gardner et al. (1988)	182	251	Y	Y	N	Holstein	Intensive	Growth rate	42	Calving	738, 666	72
Hoffman et al. (1996)-1	17	18	Y	Y	N	Holstein	Intensive	Postpubertal growth and conception rate	300	End of breeding	768, 618	150
Hoffman et al. (1996)-2	17	18	Y	Y	N	Holstein	Intensive	Postpubertal growth and conception rate	300	End of breeding	768, 681	87
Hoffman et al. (1996)-3	17	17	Y	Y	N	Holstein	Intensive	Postpubertal growth and conception rate	300	End of breeding	768, 708	60
Ishii et al. (2011)-1	11	9	Y	Y	N	Holstein	Intensive	ADG and feed protein levels	90	350 kg	693, 630	63
Ishii et al. (2011)-2	11	20	Y	Y	N	Holstein	Intensive	ADG and feed protein levels	90	350 kg	693, 654	39
Lacasse et al. (1993)-1	15	15	Y	Y	N	Holstein	Intensive	Feeding levels before and after calving	365	Calving	753, 747 ⁴	6
Lacasse et al. (1993)-2	15	14	Y	Y	N	Holstein	Intensive	Feeding levels before and after calving	365	Calving	753, 744 ⁴	9
Lacasse et al. (1993)-3	15	14	Y	Y	N	Holstein	Intensive	Feeding levels before and after calving	365	Calving	753, 735 ⁴	18
Le Cozler et al. (2009)-1	15	33	N	N	Y	Holstein	Mixed	Age at first calving, grass-fed	120	360	1,083, 728	355
Le Cozler et al. (2009)-2	12	19	N	N	Y	Holstein	Mixed	Age at first calving, corn silage low grain	120	360	1,084, 724	360
Le Cozler et al. (2009)-3	7	23	N	N	Y	Holstein	Mixed	Age at first calving, corn silage high grain	120	360	1,070, 724	346
Lin et al. (1986, 1988) ⁶	249	253	Y	Y	N	Holstein, Ayrshire, and X	Intensive	Breeding start date	Breeding	Calving	796, 698	98
Peri et al. (1993)-1	5	5	Y	Y	Y	Holstein	Intensive	Feeding level prepuberty and postpuberty	170	343	729, 700 ⁵	29

Continued

Table 1 (Continued). Summary of descriptive details of each experiment included in the meta-analysis on the intervention of age at first calving (AFC)¹

Experiment	No. of animals			Breed	Pseudoreplicated	Randomized	Housing system ²	Intervention	Intervention			AFC difference (d)
	C	T							Start age ³ (d)	Finish ³ (d)	AFC ^{4,5}	
Peri et al. (1993)-2	5	5	Y	Holstein	Y		Intensive	Feeding level prepuberty and postpuberty	170	343	729, 667 ⁵	62
Prado et al. (2021)-1	50	60	Y	Holstein	N		Intensive	Calf housing, social vs. individual housing	14	61	684, 669	15
Prado et al. (2021)-2	50	100	Y	Holstein	N		Intensive	Calf housing, social vs. individual housing	14	61	684, 663	21
Radcliff et al. (2000)-1	35	35	Y	Holstein	Y		Intensive	Growth rate	90–120	Pregnancy	719, 631	88
Radcliff et al. (2000)-2	35	35	Y	Holstein	Y		Intensive	Growth rate	90–120	Pregnancy	719, 622	97
Raeth-Knight et al. (2009)-1	54	26	Y	Holstein	N		Intensive	Milk replacer volume and solid levels	3	42	745, 731	14
Raeth-Knight et al. (2009)-2	54	29	Y	Holstein	N		Intensive	Milk replacer volume and solid levels	3	42	745, 738	7
Raeth-Knight et al. (2009)-3	54	24	Y	Holstein	N		Intensive	Milk replacer volume and solid levels	3	42	745, 717	28
Terré et al. (2009)	31	29	Y	Holstein	N		Intensive	Conventional vs. intensive diet	10	72	728, 720 ⁴	8
Van Amburgh et al. (1998)-1	97	77	Y	Holstein	N		Intensive	Prepubertal growth rates	90 ³	320 ³	735, 660	75
Van Amburgh et al. (1998)-2	97	97	Y	Holstein	N		Intensive	Prepubertal growth rates	90 ³	320 ³	735, 639	96

¹Details include authors (experiment number is listed after the year), the numbers of animals enrolled in the experiment, whether the experiment was randomized, whether the experiment was pseudoreplicated, breed of heifers, housing systems or feed systems used for the stock, experimental intervention leading to AFC difference, start and finish of the intervention, raw mean AFC control followed by treatment and the difference in AFC age. C = control; T = treatment; Y = yes; N = no; HF = Holstein-Friesian.

²Mixed housing system is a combination of pasture and intensive feeding programs.

³Start and finish is a weight in kilograms.

⁴Converted from months to days by multiplying by 30.

⁵Converted from age at conception by adding 280 d.

⁶Lin et al. (1986, 1988) combined as studies involve the same animals.

Table 2. Production outcomes by age at first calving intervention group for the experiments used in the meta-analysis¹

Experiment ²	FLMY (kg)		SLMY (kg)		TLMY (kg)		First-lactation milk fat (kg)		First-lactation milk protein (kg)	
	Control	Treatment	Control	Treatment	Control	Treatment	Control	Treatment	Control	Treatment
Bar-Peled et al. (1997)	9,171 ± 1,145 ³	9,624 ± 1,448 ³								
Davis Rincker et al. (2011)	9,778 ± 1,007	10,069 ± 1,057								
Dobos et al. (2001)-1	5,198 ± 566	4,341 ± 430								
Dobos et al. (2001)-2	5,198 ± 566	4,835 ± 846								
Dobos et al. (2001)-3	5,340 ± 531	4,825 ± 536								
Dobos et al. (2001)-4	5,340 ± 531	5,180 ± 680								
Dobos et al. (2001)-5	5,471 ± 910	4,843 ± 583								
Dobos et al. (2001)-6	5,471 ± 910	4,939 ± 609								
Dobos et al. (2004)-1	Reported in Dobos et al. (2001)		5,512 ± 1,074	4,975 ± 798	5,383 ± 834	5,346 ± 593				
Dobos et al. (2004)-2			5,512 ± 1,074	5,518 ± 876	5,383 ± 834	5,766 ± 770				
Gardner et al. (1988)			7,790 ± 1,662	7,842 ± 1,623	8,200 ± 1,888	8,330 ± 1,756				
Hoffman et al. (1996)-1	6,985 ± 1,390	6,729 ± 1,299								
Hoffman et al. (1996)-2	8,070 ± 1,390	7,522 ± 1,431								
Hoffman et al. (1996)-3	8,070 ± 1,390	7,868 ± 1,431								
Hoffman et al. (1996)-3	8,070 ± 1,390	8,291 ± 1,390								
Ishii et al. (2011)-1	8,583 ± 1,267	5,940 ± 1,452								
Ishii et al. (2011)-2	8,583 ± 1,267	6,938 ± 1,149								
Le Cozler et al. (2009)-1	4,959 ± 2,062 ⁴	4,651 ± 2,062 ⁴								
Le Cozler et al. (2009)-2	6,417 ± 2,062 ⁴	4,522 ± 2,062 ⁴								
Le Cozler et al. (2009)-3	5,787 ± 2,062 ⁴	4,119 ± 2,062 ⁴								
Lin et al. (1986, 1988)	4,452 ± 1,622	4,177 ± 1,459								
Peri et al. (1993)-1	7,056 ± 597 ⁵	6,070 ± 695 ⁵	4,795 ± 3,660	4,751 ± 3,509	6,161 ± 3,243	6,162 ± 3,006				
Peri et al. (1993)-2	7,056 ± 597 ⁵	5,975 ± 968 ⁵								
Prado et al. (2021)-1	10,537 ± 3,613	10,729 ± 3,501	10,964 ± 5,501 ⁶	10,943 ± 5,283 ⁶	8,358 ± 8,817 ⁶	7,645 ± 8,575 ⁶				
Prado et al. (2021)-2	10,537 ± 3,613	9,976 ± 3,900	10,964 ± 5,501 ⁶	9,907 ± 5,260 ⁶	8,358 ± 8,817 ⁶	6,875 ± 8,550 ⁶				
Radcliff et al. (2000)-1	8,618 ± 1,467	7,503 ± 1,467								
Radcliff et al. (2000)-2	8,618 ± 1,467	7,982 ± 1,366								
Raeth-Knight et al. (2009)-1	12,962 ± 2,292	13,123 ± 1,668								
Raeth-Knight et al. (2009)-2	12,962 ± 2,292	12,769 ± 1,801								
Raeth-Knight et al. (2009)-3	12,962 ± 2,292	13,680 ± 1,844								
Terré et al. (2009)	9,888 ± 3,913	10,512 ± 3,913								
Van Amburgh et al. (1998)-1	9,873 ± 1,228	9,620 ± 1,080	11,030 ± 820	10,940 ± 761						
Van Amburgh et al. (1998)-2	9,873 ± 1,228	9,387 ± 1,235	11,030 ± 820	11,116 ± 899						

¹First-lactation milk yield (FLMY), second-lactation milk yield (SLMY), third-lactation milk yield (TLMY), and first-lactation milk fat and protein containing raw means ± SD. Control = late-calving heifers; treatment = early-calving heifers.

²Experiment number is listed after the year.

³300-d milk yield.

⁴252-d milk yield.

⁵250-d milk yield.

⁶Second- and third-lactation milk yield data are the average production of animals enrolled in the trial from the start.

tervention start and finish, and AFC difference. The data were reported as mean and SE with the SD calculated and recorded. Values presented in figures only were extracted using image extraction software (<https://automeris.io/WebPlotDigitizer/>).

Data extracted for each treatment and control group within the experiments included the number of cows at enrollment, at calving, and after each completed lactation; first-lactation milk yield (kg); first-lactation fat yield (kg); first-lactation protein yield (kg); second-lactation milk yield (kg); third-lactation milk yield (kg); AFC; ADG during treatment time (kg/d); weight at calving (kg); calving difficulty (score 1 to 5); and heifer reproductive performance (Tables 2 and 3).

Survival data were calculated using the heifer numbers from enrollment, at calving, and at the end of each lactation. Survival was calculated using

$$(\text{number of animals at time point } x)$$

$$/(\text{number of animals enrolled}),$$

where x represents either calving or the end of first lactation.

Other data that were initially identified for collection and analysis included CP in the heifer premating diet, reproductive performance in first and second lactation, BCS at calving, second- and third-lactation fat and

Table 3. Raw means $\pm$ SD of calving difficulty, heifer reproductive performance, and survival data for early-calving (treatment) and late-calving (control) heifers

Experiment ¹	Calving difficulty (score 1–5)		Heifer reproductive performance (services per pregnancy)		Survival after calving (%)		Survival to first lactation (%)	
	Control	Treatment	Control	Treatment	Control	Treatment	Control	Treatment
Bar-Peled et al. (1997)			74.2 $\pm$ 4 ²	83.4 $\pm$ 5 ²	70	75	70	75
Davis Rincker et al. (2011)	2.1 $\pm$ 1.1	2.1 $\pm$ 1.15	1.2 $\pm$ 0.5	1.2 $\pm$ 0.6	80	83	73	80
Dobos et al. (2001)-1					93	93	87	73
Dobos et al. (2001)-2					93	87	87	87
Dobos et al. (2001)-3					80	87	80	80
Dobos et al. (2001)-4					80	87	80	87
Dobos et al. (2001)-5					93	80	87	80
Dobos et al. (2001)-6					93	87	87	80
Dobos et al. (2004)-1					89	78	84	78
Dobos et al. (2004)-2					89	84	84	84
Gardner et al. (1988)								
Hoffman et al. (1996)-1	3.4 $\pm$ 1.6	2.7 $\pm$ 1.61			100	100	100	100
Hoffman et al. (1996)-2	3.4 $\pm$ 1.6	4.2 $\pm$ 1.61	2.0 $\pm$ 0.8	1.9 $\pm$ 0.9	100	100	100	100
Hoffman et al. (1996)-3	3.4 $\pm$ 1.6	2.8 $\pm$ 1.57			100	100	100	100
Ishii et al. (2011)-1	2.5 $\pm$ 1.3	2.4 $\pm$ 1.50	2.1 $\pm$ 1.7	1.9 $\pm$ 1.8	100	100	100	100
Ishii et al. (2011)-2	2.5 $\pm$ 1.3	2.2 $\pm$ 0.89	2.1 $\pm$ 1.7	2.0 $\pm$ 1.3	100	100	100	100
Lacasse et al. (1993)-1			1.4 $\pm$ 0.7	1.6 $\pm$ 0.7	93	80	87	60
Lacasse et al. (1993)-2			1.4 $\pm$ 0.8	1.6 $\pm$ 0.7	93	93	87	79
Lacasse et al. (1993)-3			1.4 $\pm$ 0.8	1.4 $\pm$ 0.7	93	87	87	73
Le Cozler et al. (2009)-1			71 $\pm$ 37.1 ³	55 $\pm$ 60.9 ³				
Le Cozler et al. (2009)-2			44 $\pm$ 33.2 ³	75 $\pm$ 46.2 ³				
Le Cozler et al. (2009)-3			43 $\pm$ 28.1 ³	79 $\pm$ 70.5 ³				
Lin et al. (1986, 1988)	25.0 $\pm$ 46 ⁴	16 $\pm$ 45 ⁴	0.5 $\pm$ 0.8 ⁵	0.4 $\pm$ 0.8 ⁵	90	85	65	69
Peri et al. (1993)-1					100	100	100	100
Peri et al. (1993)-2					100	100	100	100
Prado et al. (2021)-1					96	95		
Prado et al. (2021)-2					96	92		
Radcliff et al. (2000)-1	2.4 $\pm$ 1.7	2.6 $\pm$ 1.7	1.4 $\pm$ 1.8	1.8 $\pm$ 1.8	91	89	86	86
Radcliff et al. (2000)-2	2.4 $\pm$ 1.7	2.0 $\pm$ 1.6	1.4 $\pm$ 1.8	2.2 $\pm$ 1.8	91	77	86	74
Raeth-Knight et al. (2009)-1							63 ⁶	69 ⁶
Raeth-Knight et al. (2009)-2							63 ⁶	72 ⁶
Raeth-Knight et al. (2009)-3							63 ⁶	92 ⁶
Terré et al. (2009)							45	48
Van Amburgh et al. (1998)-1					87	84	87	84
Van Amburgh et al. (1998)-2					87	88	87	88

¹Experiment number is listed after the year.

²Conception rate.

³Fertility percentage.

⁴The percentage cattle that had calving difficulties.

⁵Conception rate at first service.

⁶Nine heifers eliminated due to lack of milk data rather than exits for production or health.

protein yield, health outcomes, and lifetime survival. Insufficient data for these outcomes were present in the available experiments.

Statistical Analysis

Data exploration was performed using Stata (Version 18, StataCorp. LP, College Station, TX). The Stata “summarize” function was used to investigate the number of observations, mean, minimum, maximum, and SD. Histograms were used to visualize the distribution and normality of the data for each outcome.

The outcomes considered to have sufficient studies and data for analysis were first-lactation milk yield (FLMY), second-lactation milk yield (SLMY), third-lactation milk yield (TLMY), first-lactation milk fat, first-lactation milk protein, calving difficulty, and heifer reproductive performance.

Continuous variables with consistently reported units (FLMY, milk fat, milk protein) were analyzed by standard mean difference (SMD) using Stata “esize(mdif).” The SMD variables were pooled using DerSimonian and Laird (1986) random effects model. Only random effects models were used, as previous work concluded that when there was uncertainty in the evaluative units caused by clustering of observations, the random effects model was appropriate (White and Thomas, 2005). When the outcomes were measured with variable units (calving difficulty, reproductive performance, and multiparous milking performance), effect size (ES) was used in the model Stata “esize(cohend).” The SLMY and TLMY were initially to be analyzed as SMD, with kilograms per lactation per animal as the units; however, Prado et al. (2021) reported their milk production findings as an average volume of all heifers enrolled in the experiment, including those that had exited the experiment. Therefore, ES was used to analyze these outcomes.

Survival data were the only dichotomous variable in the dataset and were evaluated using the Stata “esize(rdif)” function. As there were experiments nested within studies for all variables, Stata “meta multilevel” was used to provide a better measure of SMD, ES, and risk difference (Goldstein et al., 2000; Thompson et al., 2001). The es-

timates of SMD, ES, or risk difference for each outcome were presented using Stata “meta forestplot.”

Variations among the experimental comparison level SMD were assessed using a chi-squared (Q) test of heterogeneity. Heterogeneity in experimental comparisons reflects underlying differences in clinical diversity of the research site and interventions, differences in experimental design and analytical methods, and statistical variation around responses. The clinical diversity of the site includes all the nonstudy design aspects of variation, such as facility design, environment, feed systems, and cattle management, that may be measured and controlled for in meta-analysis but are often not reported or measured. Identifying the presence and sources of heterogeneity improves understanding of the responses to the interventions used. An α level of 0.1 was used because of the relatively poor power of the χ^2 test to detect heterogeneity among small numbers of trials (Egger and Smith, 2001). Heterogeneity of results among the studies and also experimental comparisons were quantified using the I^2 statistic (Higgins and Thompson, 2002) and the H^2 statistic (Lin et al., 2017) and were reported as the total estimates from the study and experiment levels. The I^2 provides an estimate of the proportion of the true variance of effects of the treatment, that is the true variance, tau-squared (τ^2) divided by the total variance observed in the comparison (Borenstein et al., 2017), which reflects measurement error. The H^2 provides an estimate of heterogeneity and where H is interpreted as the ratio of the SD of the estimated overall ES from a random effects meta-analysis compared with the SD from a fixed effects meta-analysis, in which an H^2 of 1 indicates no heterogeneity (Lin et al., 2017). This multilevel analysis used the total study I^2 Higgins-Thompson estimation using Stata “estat heterogeneity” based on methods developed by Nakagawa and Santos (2012). While estat heterogeneity provides estimates of heterogeneity at the study and experiment level, we report the total heterogeneity only. The univariate I^2 were assessed but are not reported. An I^2 value between 0% and 40% might not be important, 30% to 60% may represent moderate heterogeneity, 50% to 90% might represent substantial heterogeneity, and 75% to 100% might represent considerable heterogeneity (Higgins and Green, 2011). We provide I^2 , τ^2 , and H^2 to

Table 4. Summary of the standard mean difference and heterogeneity estimates for first-lactation milk production outcomes for age at first calving

Outcome (kg/lactation)	No. of studies	No. of experiments	Standard mean difference (95% CI)	I^2 estimated heterogeneity (%)	τ^2	H^2 estimate	P-value
First-lactation milk yield	14	30	-430.119 (-819.356, -40.881)	100	639,572	1,812	0.030
First-lactation milk fat	8	19	-17.507 (-31.856, -3.158)	98	565	45	0.017
First-lactation milk protein	8	19	-14.740 (-25.837, -3.643)	97	312	31	0.009

Table 5. Summary of effect size differences and heterogeneity estimates for calving difficulty, multiparous milk yield, and heifer reproductive performance for age at first calving

Outcome	No. of studies	No. of experiments	Effect size (95% CI)	I^2 estimated heterogeneity (%)	τ^2	H^2 estimate	P -value
Calving difficulty	5	9	-0.121 (-0.450, 0.210)	0	0	1	0.469
Second-lactation milk yield	5	8	0.08 (-0.42, 0.57)	0	0	1	0.760
Third-lactation milk yield	4	6	0.05 (-0.59, 0.50)	0	0	1	0.864
Heifer reproductive performance	8	14	-0.032 (-0.308, 0.244)	0	0	1	0.818

allow readers the opportunity to evaluate those metrics. The dichotomized data were also evaluated for heterogeneity using L'Abbe plots (Stata “meta labbeplot”; data not shown).

The datasets were evaluated for publication bias with the meta “funnelplot” function. Funnel plots are a scatter plot of the intervention effect estimates from individual comparisons plotted against comparison precision. The name “funnel plot” arises because the precision of the intervention effect increases as the size and precision of a comparison increases. Effect estimates from comparisons with a small number of animal units will scatter more widely at the bottom of the graph, and the spread narrows for those with increasing numbers of units. In the absence of bias, the plot should approximately resemble a symmetrical (inverted) funnel. Data with more than 3 experiments outside the funnel plot area were evaluated using the “meta bias, egger” function to evaluate for small study bias, although this was not found to be present in this study (data not shown).

The risk of bias assessment within each study was considered using the guidelines set out by Winder et al. (2019). Randomized studies were compared with pseudorandomized studies within the models, but no difference was found. There was a risk of bias due to the lack of blinding in all of the studies used in this analysis, as well as a limited number of outcomes with clear experimental numbers analyzed. We considered the risk of bias assessment due to these problems to be present but of low impact.

All outcome variables were investigated for possible covariates and confounding in multilevel models. Vari-

ables tested included the difference in days between the AFC, weight at calving, probable pseudoreplication, and ADG. The ADG difference was taken from the period of intervention. Meta-regression was also performed for all outcomes using the Stata “meta regress” function for the AFC difference, and confounding was investigated using the covariates mentioned above. These data were visualized and presented in bubble plots where significance was observed. If a coefficient changed by more than 25% in this analysis, then a covariate would be considered to be confounding if the number of studies analyzed was not changed (Rothman, 1986). Stata “meta summarize leaveoneout” was used to assess the influence of a single experiment on results. No single experiment was seen to confound the results in any of the analyses.

RESULTS

The meta-analysis results are summarized in Tables 4, 5, and 6, and individual outcomes are displayed in the forest plots (Figures 2, 3, 4, 5, 6, 7, 8, 9, and 10). Meta-regression data are summarized in Table 7, and for FLMY, first lactation milk protein, and first-lactation milk fat bubble plots are presented in Figures 11, 12, and 13.

There were limited prospective reports studying AFC that were amenable to quantitative review. There were 16 usable studies, with 35 experiments (Table 1). Experiments with larger numbers of heifers per experimental group were from older studies, with the 2 largest studies (>180 heifers per experimental group) being from the 1980s. There was no indication of bias from the funnel

Table 6. Summary of risk difference for survival data for cattle that have a younger age at first calving

Outcome	No. of studies	No. of experiments	Risk difference (95% CI)	I^2 estimated heterogeneity (%)	τ^2	H^2 estimate	P -value
Survival to calving	11	25	-0.01 (-0.20, 0.19)	0	0	1	0.954
Survival to first lactation	12	27	0.01 (-0.17, 0.20)	0	0	1	0.905

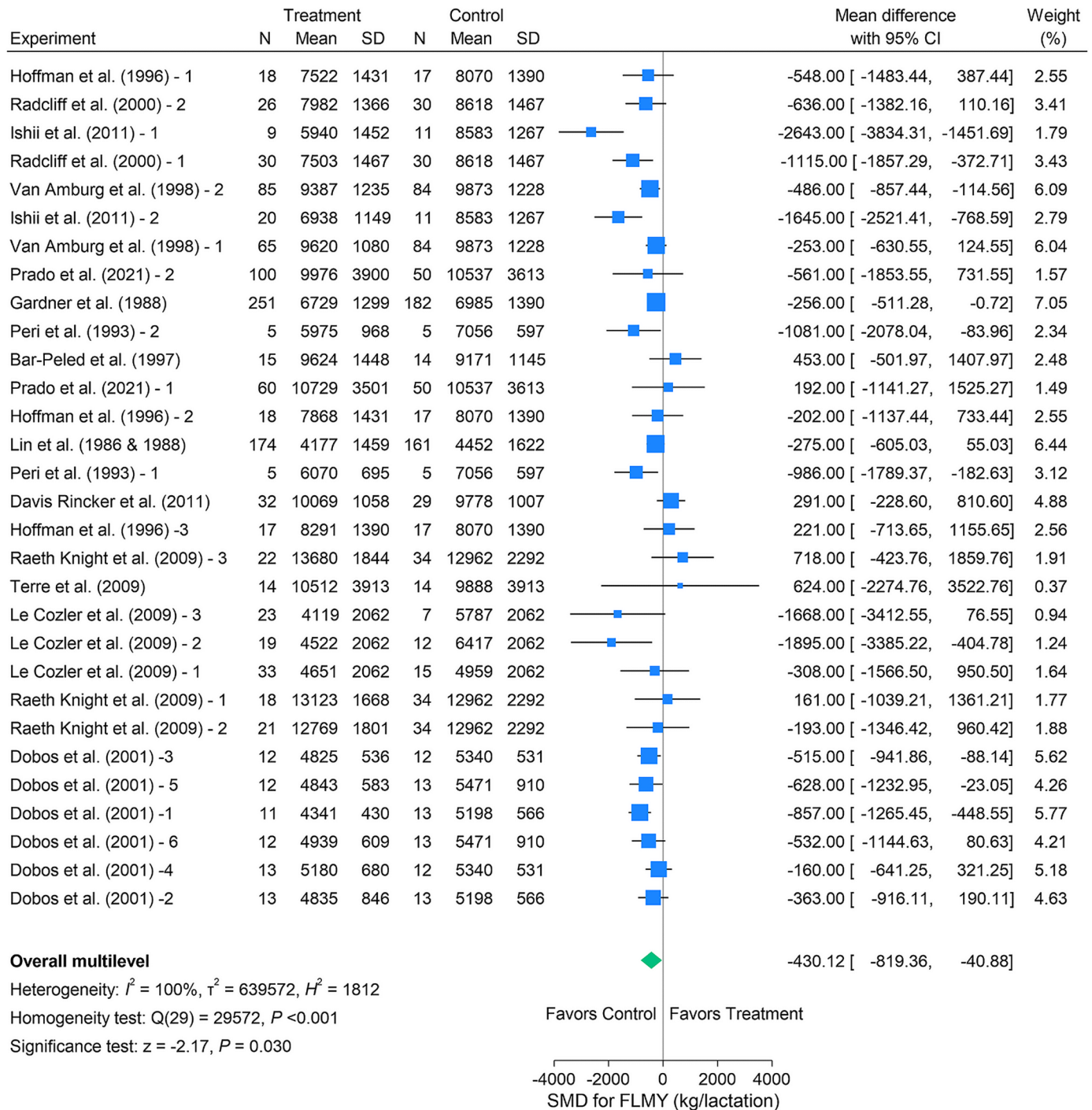


Figure 2. A forest plot of the standard mean difference (SMD) and 95% CI for first-lactation milk yield (FLMY). Standard mean difference is the mean difference between the group standardized to kilograms/lactation. The SMD of the whole group is the calculated mean difference of the whole group after multilevel meta-analysis. The solid vertical line represents no effect or zero. Points to the left of the line indicate a reduction in milk production, whereas points to the right indicate increased milk production. Each square on the graph represents the mean effect size for comparison at the univariate level, with the lines extending from the square being the 95% CI of that experiment. The size of the square indicates the weighting of the experiment at a univariate level, the value of which is recorded in the right-hand column. The diamond at the base represents the results from the overall multilevel model, which significantly ($P = 0.030$) favors the control group. The forest plot is ordered by ascending age at first calving (AFC) of the treatment group from the top. There are high levels of heterogeneity between the experiments, as can be seen by the high values of I^2 , τ^2 , and H^2 . References: Hoffman et al. (1996), Radcliff et al. (2000), Ishii et al. (2011), Van Amburgh et al. (1998), Prado et al. (2021), Gardner et al. (1988), Peri et al. (1993), Bar-Peled et al. (1997), Lin et al. (1986, 1988), Davis Rincker et al. (2011), Raeth-Knight et al. (2009), Terré et al. (2009), Le Cozler et al. (2009), Dobos et al. (2001).

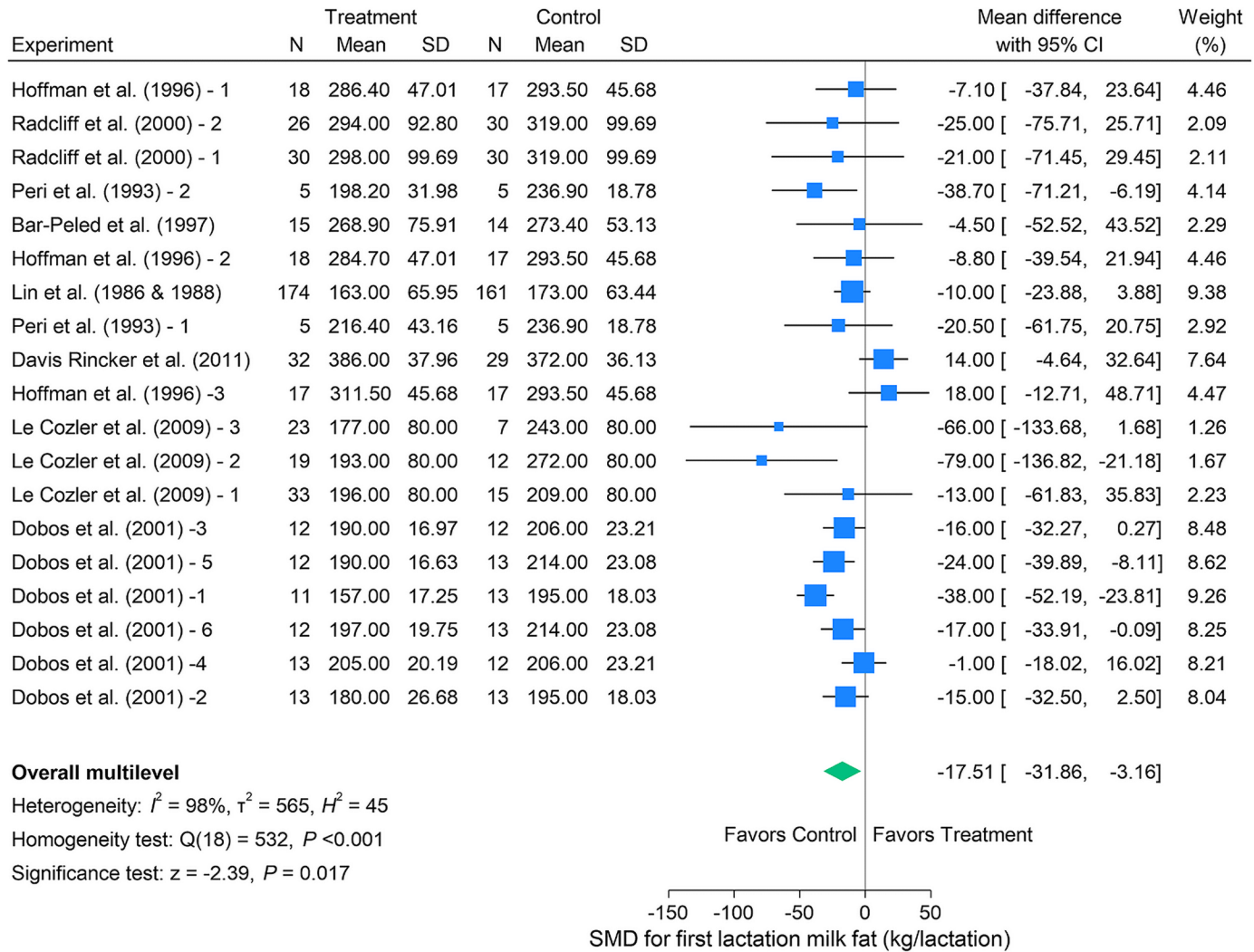


Figure 3. A forest plot of the standard mean difference for first-lactation milk fat (kg/lactation). The solid vertical line represents no effect or zero. Points to the left of the line indicate a reduction in milk fat production, whereas points to the right indicate increased milk fat production. Each square on the graph represents the mean effect size for comparison at the univariate level, with the lines extending from the square being the 95% CI of that experiment. The size of the square indicates the weighting of the experiment at a univariate level, the value of which is recorded in the right-hand column. The diamond at the base represents the results from the overall multilevel model, which significantly ($P = 0.017$) favors the control group. The forest plot is ordered by ascending age at first calving (AFC) of the treatment group from the top. There are high levels of heterogeneity between the experiments, as can be seen by the high values of I^2 , τ^2 , and H^2 . References: Hoffman et al. (1996), Radcliff et al. (2000), Peri et al. (1993), Bar-Peled et al. (1997), Lin et al. (1986, 1988), Davis Rincker et al. (2011), Le Cozler et al. (2009), Dobos et al. (2001).

plots and the bias calculations (Supplemental Figure S1, see Notes).

A range of interventions was used to facilitate a difference in AFC (Table 1). Interventions fell into 3 main categories: heifer growth rate management through variable feeding strategies ($n = 12$), age at first breeding ($n = 1$), and preweaning calf feeding methods ($n = 8$), or a combination of age at first breeding and variable feeding ($n = 14$). Experiments ranged from 10 to 502 enrolled heifers. The predominant breed used in these studies was Holstein, and most experiments used intensive cattle housing and feeding systems at some point during the

rearing period. Only one study was pasture-based, but it contributed 8 experiments (Dobos et al., 2001, 2004). The mean AFC for treatment groups ranged from 618 to 918 d while the control groups ranged from 684 to 1,084 d. The range of AFC differences was between 6 and 360 d with a mean difference of 108 d across the 35 experiments.

The FLMY was the most reported outcome variable among experiments, with all 35 experiments from 16 studies reporting this metric. One study (Lacasse et al., 1993) containing 3 experiments was not included in the final analysis due to reporting liters of milk produced

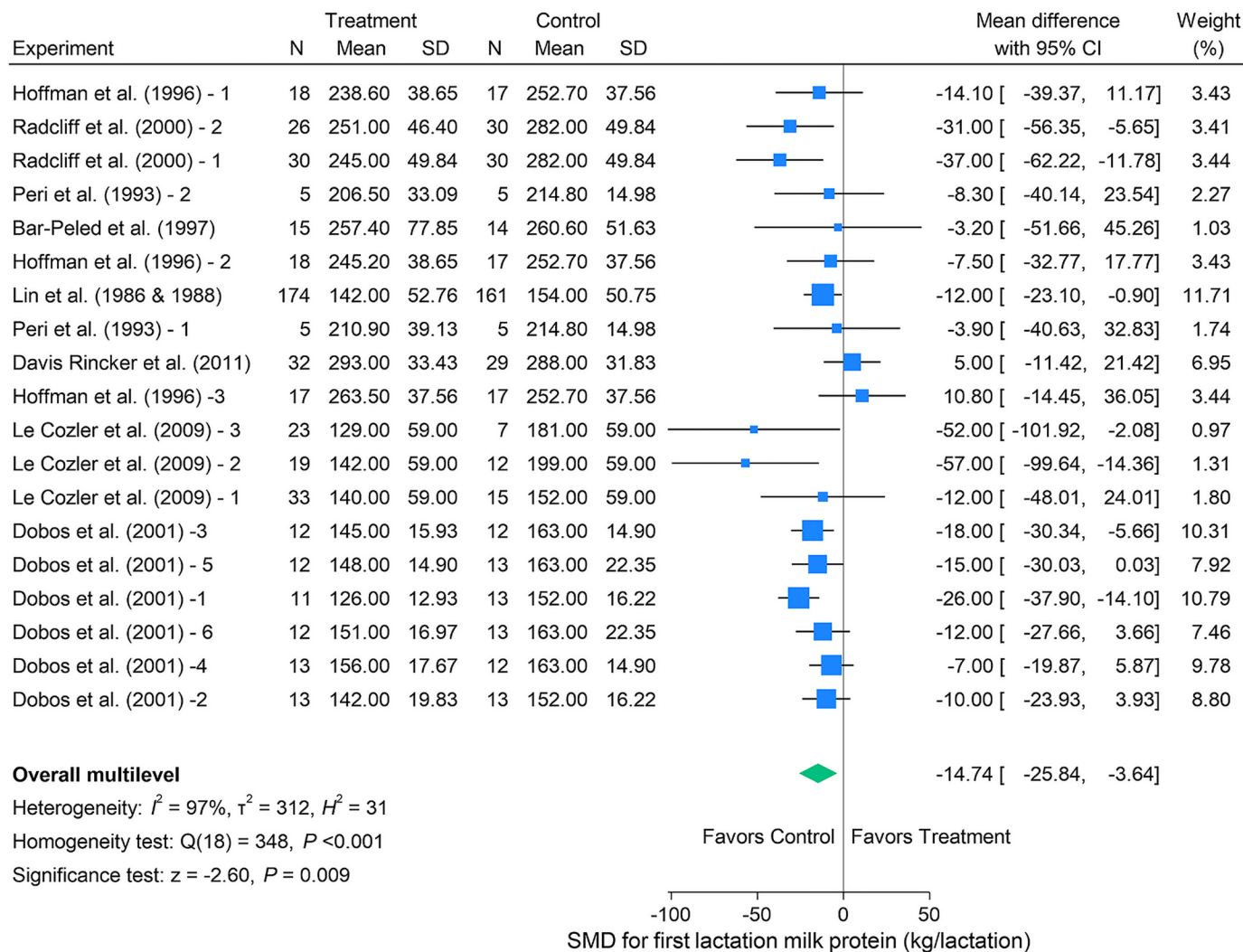


Figure 4. A forest plot of the standard mean difference for first-lactation milk protein (kg/lactation). The solid vertical line represents no effect or zero. Points to the left of the line indicate a reduction in milk protein production, whereas points to the right indicate increased milk protein production. Each square on the graph represents the mean effect size for comparison at the univariate level, with the lines extending from the square being the 95% CI of that experiment. The size of the square indicates the weighting of the experiment at a univariate level, the value of which is recorded in the right-hand column. The diamond at the base represents the results from the overall multilevel model, which significantly ($P = 0.009$) favors the control group. The forest plot is ordered by ascending age at first calving (AFC) of the treatment group from the top. There are high levels of heterogeneity between the experiments, as can be seen by the high values of I^2 , τ^2 , and H^2 . References: Hoffman et al. (1996), Radcliff et al. (2000), Peri et al. (1993), Bar-Peled et al. (1997), Lin et al. (1986, 1988), Davis Rincker et al. (2011), Le Cozler et al. (2009), Dobos et al. (2001).

per day, not kilograms per lactation or over a lactation period (Table 2). Figure 2 shows there was a decrease in milk production ($P = 0.030$) in heifers managed to have increased growth and achieve a younger AFC, compared with controls. The I^2 of 100% along with high τ^2 and H^2 values indicated considerable heterogeneity (Table 4; Figure 2). When meta-regression was performed for the effect of age difference at first calving, there was a loss of 2.06 ± 0.95 L in milk per day earlier a heifer calves ($P < 0.001$; Table 7). The bubble plot (Figure 11) shows that the larger experiments fit well on the meta-regression line. Lower FLMY was not reflected through to second

and TLMY ($P > 0.05$; Table 5 and Figures 7 and 8, respectively).

First-lactation milk fat ($P = 0.017$) and protein ($P = 0.009$) yields were decreased in the heifers managed to have increased growth and achieve a younger AFC (Table 4; Figures 3 and 4, respectively). When milk protein and fat were meta-regressed with the difference of AFC, there was a decrease ($P < 0.001$) in production as AFC decreased (Table 7; Figures 12 and 13, respectively). Milk fat was decreased by 0.12 ± 0.032 kg/d in a heifer that calved at a younger age, while milk protein was decreased by 0.08 ± 0.028 kg/d as AFC decreased.

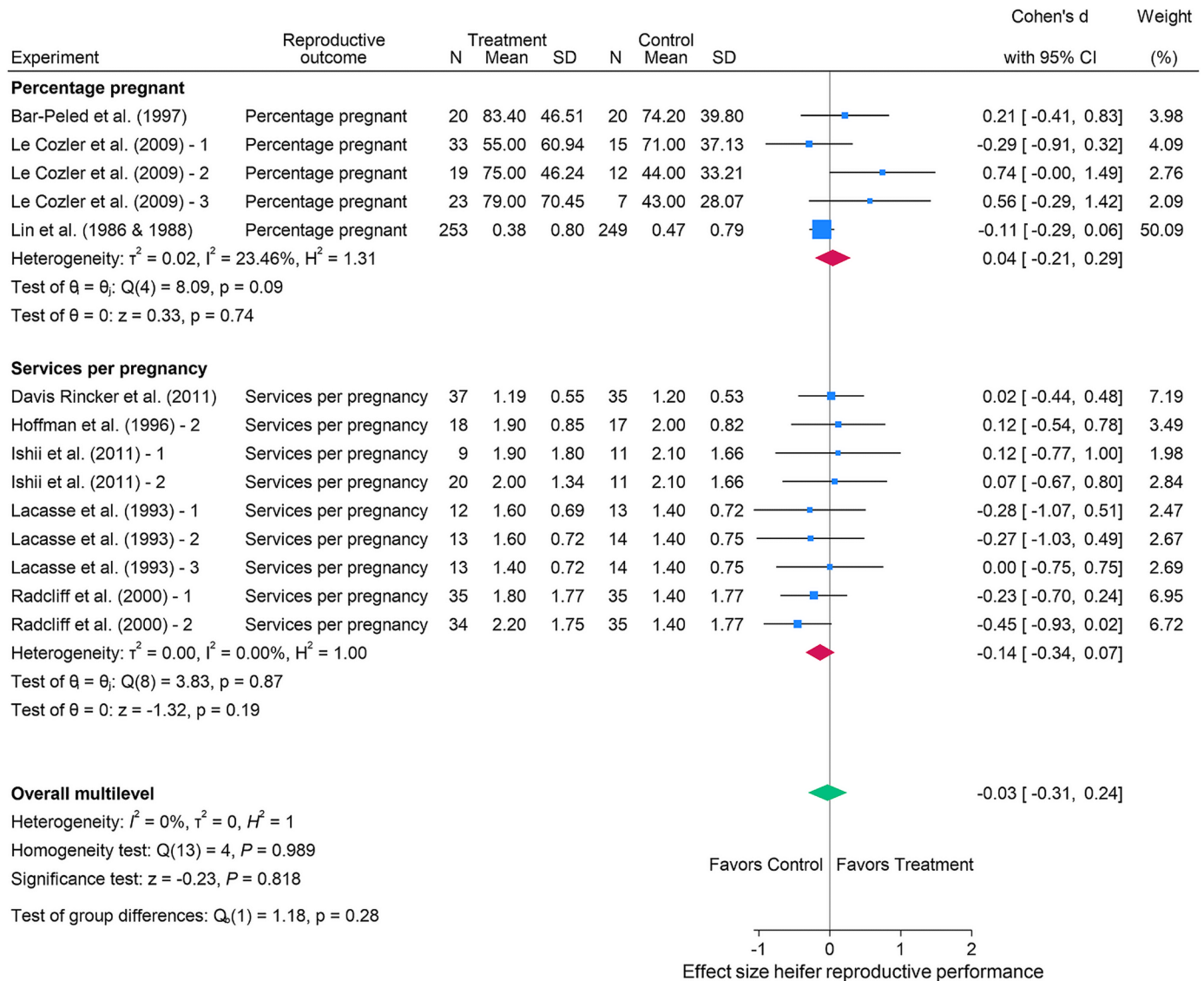


Figure 5. A forest plot representing the effect size of heifer reproductive performance split into reproductive measure subgroups. The solid vertical line represents no effect or zero. Points to the left of the line indicate a reduction in reproductive performance, whereas points to the right indicate increased reproductive performance. Each square on the graph represents the mean effect size for comparison at the univariate level, with the lines extending from the square being the 95% CI of that experiment. The size of the square indicates the weighting of the experiment at a univariate level, the value of which is recorded in the right-hand column. The red diamonds at the base of each subgroup of reproductive measures represent the results of the univariate models for those reproductive measures. The green diamond at the base represents the results from the overall multilevel model, which was nonsignificant ($P = 0.818$). The forest plot is ordered by ascending age at first calving (AFC) of the treatment group from top to bottom. There is little heterogeneity between the experiments, as can be seen by the low values of I^2 and τ^2 and high H^2 . References: Hoffman et al. (1996), Radcliff et al. (2000), Ishii et al. (2011), Bar-Peled et al. (1997), Lin et al. (1986, 1988), Davis Rincker et al. (2011), Le Cozler et al. (2009), Lacasse et al. (1993).

Therefore, a heifer calving 100 d earlier would produce 206 L less milk, 12 kg less fat, and 8 kg less protein over their first lactation. The weight at calving was not found to be confounding as a covariate in meta-regression for production outcomes (data not shown), although ADG in the intervention period was confounding for milk protein and fat yields, decreasing the amount of milk protein and fat loss as AFC decreased.

Data for nonproduction outcomes (reproductive performance, calving difficulty, and survival) were highly homogeneous (Tables 5 and 6). The H^2 values approached 1, while the I^2 values were near 0.

Data on the reproductive performance of the heifers were present in 20 experiments (Figure 8). Four experiments were removed from the final analysis, as they did not contain measures of variance ($n = 2$) or reproductive

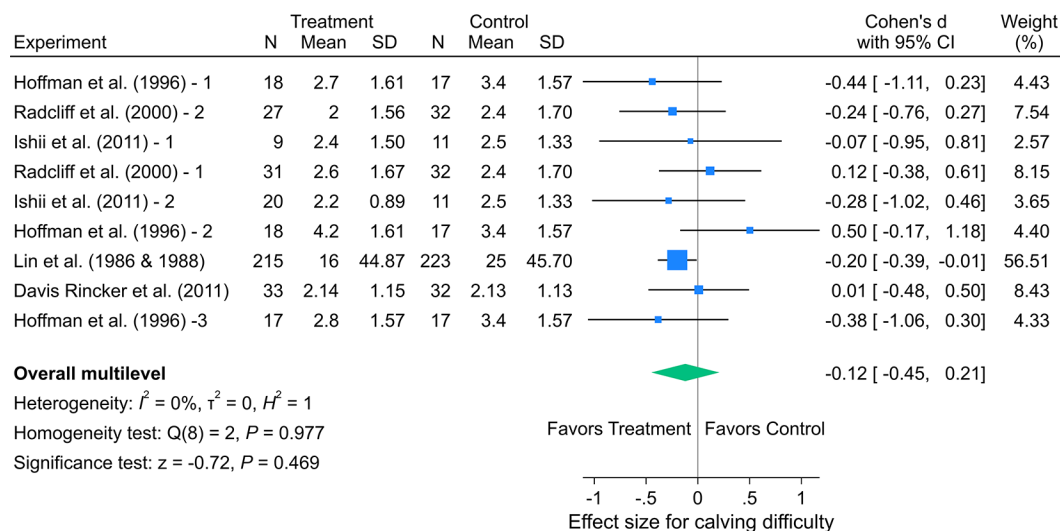


Figure 6. A forest plot representing the effect size of calving difficulty. The solid vertical line represents no effect or zero. Points to the left of the line indicate a reduction in calving difficulty, whereas points to the right indicate increased calving difficulty. Each square on the graph represents the mean effect size for comparison at the univariate level, with the lines extending from the square being the 95% CI of that experiment. The size of the square indicates the weighting of the experiment at a univariate level, the value of which is recorded in the right-hand column. The diamond at the base represents the results from the overall multilevel model, which was nonsignificant ($P = 0.469$). The forest plot is ordered by ascending age at first calving (AFC) of the treatment group from the top. There is little heterogeneity between the experiments, as can be seen by the low values of I^2 and τ^2 and high H^2 . References: Hoffman et al. (1996), Radcliff et al. (2000), Ishii et al. (2011), Lin et al. (1986, 1988), Davis Rincker et al. (2011).

performance was used to produce an AFC difference ($n = 2$). Measures used to assess reproductive performance are summarized in Table 3 but include conception rate, fertility percentage, and conception rate at first service. These were grouped into subgroups of services per pregnancy and percentage pregnant as well as providing an overall result. There was no difference ($P = 0.818$) between the reproductive performance of heifers that were managed to have a decreased AFC and those in the control group that calved at an older age (Table 5 and Figure 8). There was also no difference found in the subgroup analysis for the percentage pregnant ($P = 0.74$) and services per pregnancy ($P = 0.19$).

Calving difficulty at first calving was assessed across 9 experiments. No differences ($P = 0.469$; Table 5; Figure 6) were found for calving difficulty with differences in AFC.

Survival data were present for heifers at first calving and the end of first lactation, though data were too limited to continue the analysis into later lactations. Gardner et al. (1988) was removed from the analysis, as no information about numbers at enrollment was provided, so survival data could not be used. There was no difference between AFC groups for survival to first calving ($P = 0.954$; Table 6; Figure 9) and survival to the end of first lactation ($P = 0.905$; Figure 10), with the risk difference of the meta-analysis approaching 0 for both these outcomes. This means survival to the end of the first lactation was nearly identical regardless of AFC.

DISCUSSION

The AFC research is intended to find the most efficient AFC for the modern dairy industry, as feeding strategies and genetics have improved rapidly. While AFC is a topic that has been well researched, with a large amount of literature, the number of experiments that met the criteria for this meta-analysis was relatively low. There were many nonrandomized or retrospective, or both studies in this area of research, as well as several experiments that used an older AFC (above 26 mo) for their treatment groups. Only one suitable study was identified from the last 10 yr, meaning all the information available on AFC is older or does not come from randomized controlled trials. While only 35 experiments were eligible for meta-analysis, the areas of interest: FLMY, first-lactation milk fat and protein yield, SLMY, TLMY, calving difficulty, survival to calving, survival to the end of first lactation, and reproductive performance as a heifer were all investigated.

Other areas of interest that could not be investigated by meta-analysis were second- and third-lactation fat and protein yield, reproductive performance postcalving, lifetime survival time, and health outcomes. These areas will need further prospective studies to allow for evaluation of these outcomes. There was a paucity of studies that investigated more than one lactation ($n = 8$).

This meta-analysis is limited to recommendations involving Holstein dairy cows. One study in our dataset

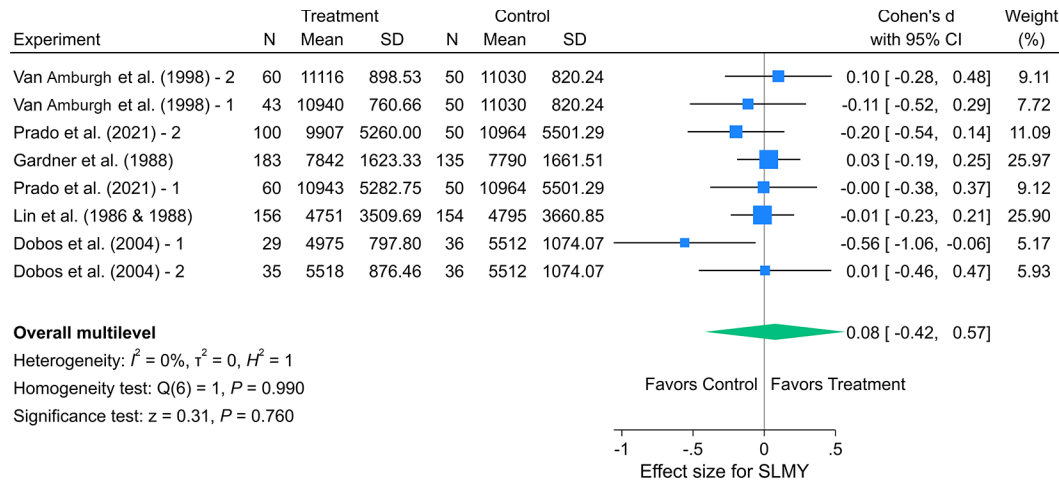


Figure 7. A forest plot representing the effect size of second-lactation milk yield (SLMY). The solid vertical line represents no effect or zero. Points to the left of the line indicate a reduction in milk production, whereas points to the right indicate increased milk production. Each square on the graph represents the mean effect size for comparison at the univariate level, with the lines extending from the square being the 95% CI of that experiment. The size of the square indicates the weighting of the experiment at a univariate level, the value of which is recorded in the right-hand column. The diamond at the base represents the results from the overall multilevel model, which was nonsignificant ($P = 0.760$). The forest plot is ordered by ascending age at first calving (AFC) of the treatment group from the top. There is little heterogeneity between the experiments, as can be seen by the low values of I^2 and τ^2 and high H^2 . References: Van Amburgh et al. (1998), Prado et al. (2021), Gardner et al. (1988), Lin et al. (1986, 1988), Dobos et al. (2001).

contained other breeds of cattle or crossbreds (Lin et al., 1986, 1988; Table 1). Other studies investigated AFC in different breeds that were included in our search criteria. These studies identified and analyzed large retrospective databases to make their recommendations for Jersey (Sherwin et al., 2016) and Brown Swiss dairy cattle (Hutchison et al., 2017). Consequently, these were not eligible for inclusion in our study.

The differences in study design, experimental differences, and AFC range (Table 1) led to high levels of heterogeneity among experiments. This was especially evident in the outcomes with the greatest number of experiments present, the first-lactation production outcomes (Table 4). When outcomes were investigated in a smaller number of experiments, the data had little heterogeneity (Tables 5 and 6).

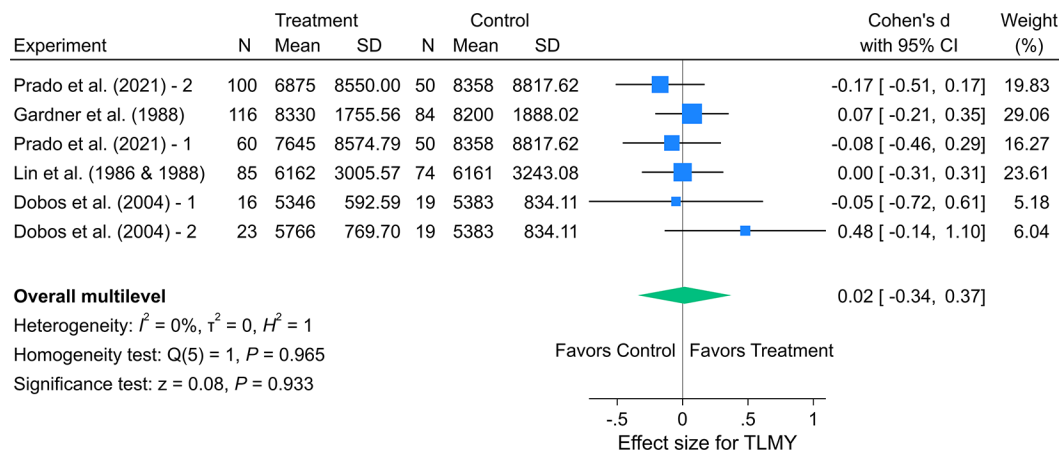


Figure 8. A forest plot representing the effect size of third-lactation milk yield (TLMY). The solid vertical line represents no effect or zero. Points to the left of the line indicate a reduction in milk production, whereas points to the right indicate increased milk production. Each square on the graph represents the mean effect size for comparison at the univariate level, with the lines extending from the square being the 95% CI of that experiment. The size of the square indicates the weighting of the experiment at a univariate level, the value of which is recorded in the right-hand column. The diamond at the base represents the results from the overall multilevel model, which was nonsignificant ($P = 0.933$). The forest plot is ordered by ascending age at first calving (AFC) of the treatment group from the top. There is little heterogeneity between the experiments, as can be seen by the low values of I^2 and τ^2 and high H^2 . References: Prado et al. (2021), Gardner et al. (1988), Lin et al. (1986, 1988), Dobos et al. (2001).

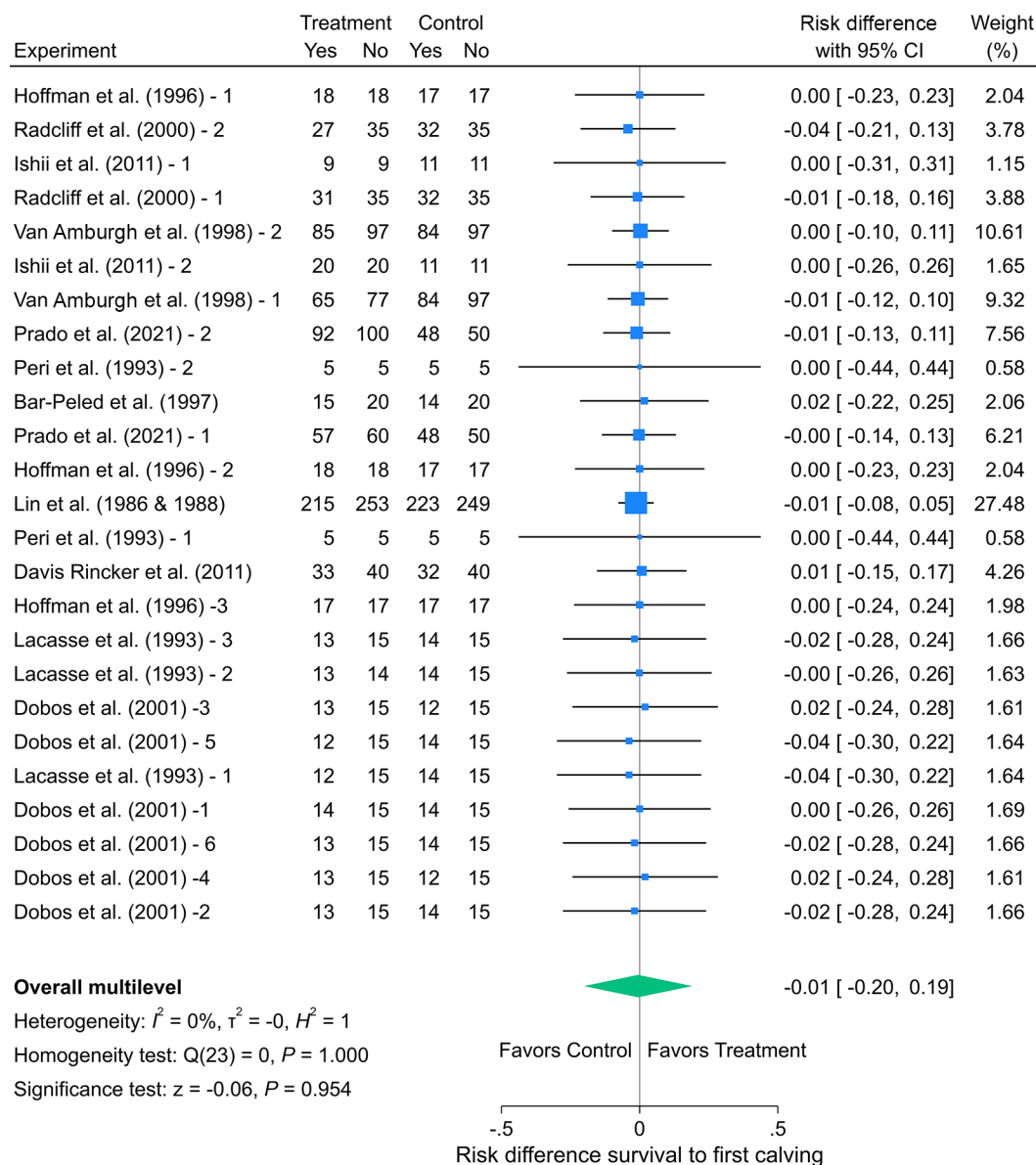


Figure 9. A forest plot representing the risk difference of survival to calving. The solid vertical line represents no effect or zero. Points to the left of the line indicate a reduction in survival to first calving, whereas points to the right indicate improved survival to first calving. Each square on the graph represents the mean effect size for comparison at the univariate level, with the lines extending from the square being the 95% CI of that experiment. The size of the square indicates the weighting of the experiment at a univariate level, the value of which is recorded in the right-hand column. The diamond at the base represents the results from the overall multilevel model, which was nonsignificant ($P = 0.954$). The forest plot is ordered by ascending age at first calving (AFC) of the treatment group from the top. There is little heterogeneity between the experiments, as can be seen by the low values of I^2 and τ^2 and high H^2 . References: Hoffman et al. (1996), Radcliff et al. (2000), Ishii et al. (2011), Van Amburgh et al. (1998), Prado et al. (2021), Peri et al. (1993), Bar-Peled et al. (1997), Lin et al. (1986, 1988), Davis Rincker et al. (2011), Lacasse et al. (1993), Dobos et al. (2001).

The decreasing FLMY over the first lactation as AFC decreased is consistent with the other studies investigating AFC, with FLMY decreasing in younger calving heifers (Ettema and Santos, 2004; Nilforooshan and Edriss, 2004; Mohd Nor et al., 2013). As is the similar performance in subsequent lactations (Krpáková et al., 2014; Hutchison et al., 2017). Krpáková et al. (2014)

reported that precalving heifers grown at an ADG >0.950 kg/d had increased lactation performance in second and third lactations compared with their older calving peers. A lack of study data meant the precalving ADG could not be evaluated as a covariate for second and third lactation in our study. There is a possibility that the relative loss of FLMY could be regained in subsequent lactations.

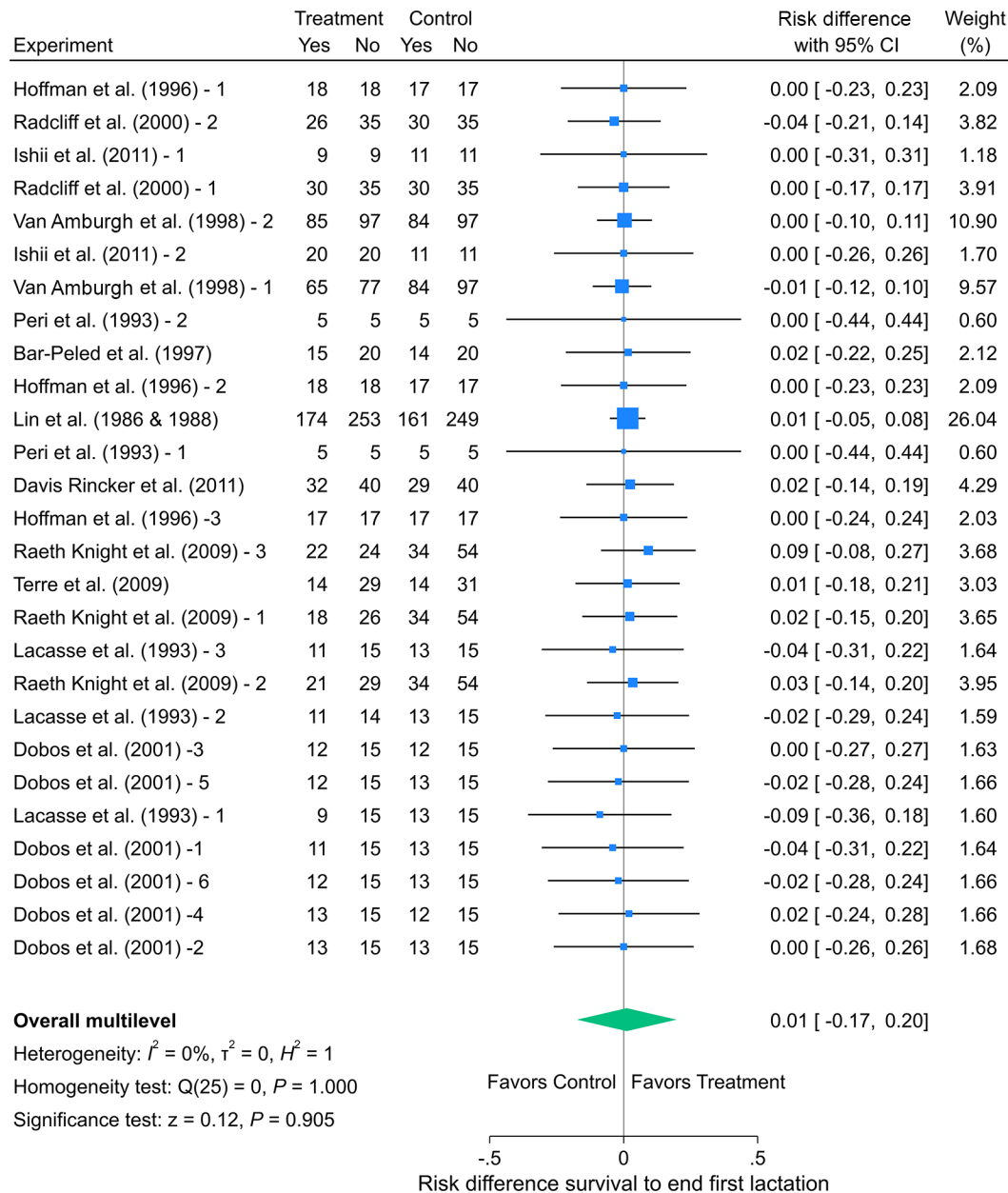


Figure 10. A forest plot representing the risk difference of survival to the end of first lactation. The solid vertical line represents no effect or zero. Points to the left of the line indicate a reduction in survival to the end of the first lactation, whereas points to the right indicate improved survival to the end of first lactation. Each square on the graph represents the mean effect size for comparison at the univariate level, with the lines extending from the square being the 95% CI of that experiment. The size of the square indicates the weighting of the experiment at a univariate level, the value of which is recorded in the right-hand column. The diamond at the base represents the results from the overall multilevel model, which was nonsignificant ($P = 0.905$). The forest plot is ordered by ascending age at first calving (AFC) of the treatment group from the top. There is little heterogeneity between the experiments, as can be seen by the low values of I^2 and τ^2 and high H^2 . References: Hoffman et al. (1996), Radcliff et al. (2000), Ishii et al. (2011), Van Amburgh et al. (1998), Peri et al. (1993), Bar-Peled et al. (1997), Lin et al. (1986, 1988), Davis Rincker et al. (2011), Raeth-Knight et al. (2009), Terré et al. (2009), Lacasse et al. (1993), Dobos et al. (2001).

The decreased lactation and solids production over the first lactation from heifers managed to have a younger AFC (0.12 kg/d and 0.08 kg/d less fat and protein with earlier calving, respectively) are potentially due to the ongoing need for body growth and development in these

younger heifers compared with their older peers. However, when weight at calving was used as a covariate, it was not confounding. While ADG in the intervention period was confounding, decreasing the amount of milk fat and protein lost, it was a method used to induce an

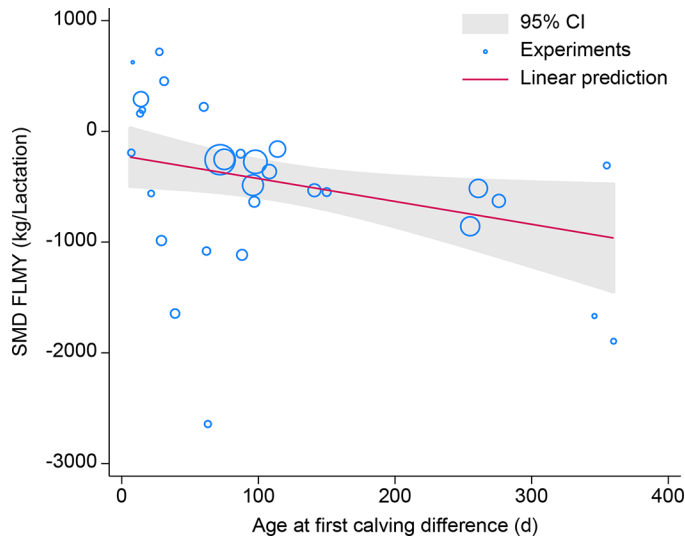


Figure 11. Bubble plot representing the standard mean difference (SMD) of first-lactation milk yield (FLMY) compared with the age at first calving difference. The experiments are represented by circles of varying sizes, with the size of the circle representing the total number of animals in the experiment.

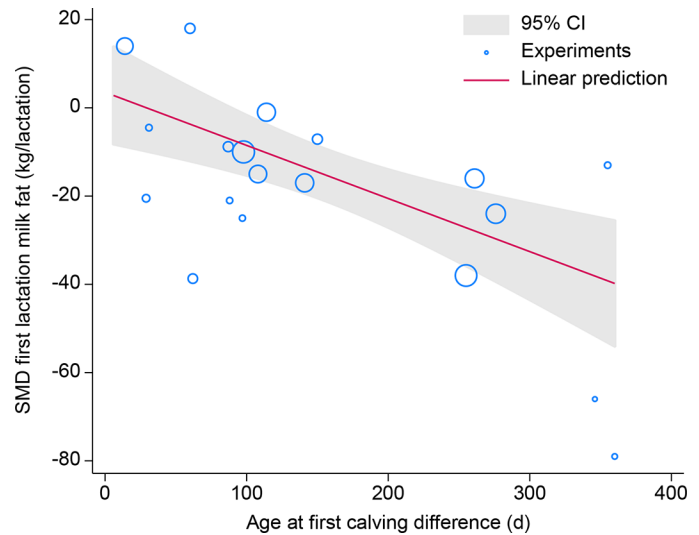


Figure 12. Bubble plot representing the standard mean difference (SMD) first-lactation milk fat (kg/lactation) compared with the AFC age difference. The studies are represented by circles of varying sizes, with the size of the circle representing the number of animals in the experiment.

AFC difference, it is therefore hard to assess whether this is a real change in milk fat and protein production or just an effect of study design.

Loss of milk and milk solids production from lowering AFC has been suggested to result from growth rates of more than 800 g/d in the prepubertal phase due to excessive fat deposition in the udder (Dobos et al., 2000; Sejrsen et al., 2000; Zanton and Heinrichs, 2005). This effect reflects increased ADG being required for heifers to reach adequate BW to conceive, calve, and produce

milk. It has been subsequently reported that it is possible to exceed these ADG without the development of excess fat deposition if there is adequate protein in the diet (Capuco et al., 1995; Dobos et al., 2000). While there were dietary data for some of the experiments, there were not enough data available to evaluate the protein content of the prepubertal diet as a covariable in this study.

There have been varied results reported in the literature for calving ease. It has been reported that lowering the

Table 7. Meta-regression effect size and constants for outcome variables for the difference in age at first calving

Outcome	No. of studies	No. of experiments	Coefficient (95% CI)	P-value	Constant (95% CI)	P-value constant
First-lactation milk yield (kg/lactation)	14	30	-2.063 (-3.927, -0.198)	0.030	-220.317 (-491.853, 51.219)	0.112
First-lactation milk fat (kg/lactation)	8	19	-0.120 (-0.183, -0.058)	<0.001	3.522 (-7.781, 14.826)	0.541
First-lactation milk protein (kg/lactation)	8	19	-0.077 (-0.123, -0.030)	0.001	-1.869 (-10.255, -6.516)	0.662
Calving difficulty	5	9	-0.002 (-0.007, 0.003) ¹	0.418	0.030 (-0.420, 0.4793)	0.896
Second-lactation milk yield	5	8	-0.001 (-0.004, 0.008) ¹	0.210	-0.070 (-0.143, 0.283)	0.520
Third-lactation milk yield	4	6	0.015 (-0.340, 0.371) ¹	0.933	0.075 (-0.318, 0.169)	0.548
Heifer reproductive performance	8	14	-0.0008 (-0.0006, 0.0023) ¹	0.259	-0.164 (-0.357, 0.029)	0.096
Survival to calving	11	25	-0.00002 (-0.001, 0.001) ²	0.947	-0.006 (-0.066, 0.054)	0.847
Survival to first lactation	12	27	-0.0001 (-0.0007, 0.0005) ²	0.667	0.0168 (-0.043, 0.076)	0.580

¹Based on Cohen's d effect size.

²Based on risk difference.

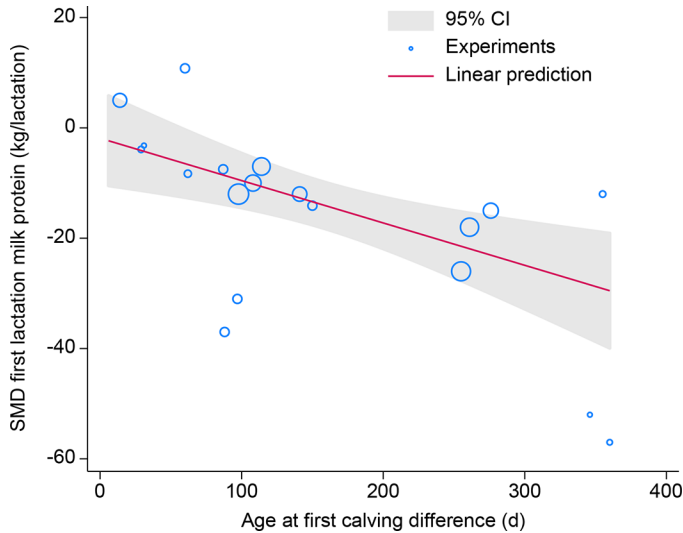


Figure 13. Bubble plot representing the standard mean difference (SMD) first-lactation milk protein (kg/lactation) compared with the age at first calving difference. The experiments are represented by circles of varying sizes, with the size of the circle representing the total number of animals in the experiment.

AFC will increase calving difficulties, as heifers will be smaller and younger and unable to deliver a calf through an underdeveloped frame as easily as older heifers (Erb et al., 1985). Erb et al. (1985) suggested that these problems could be offset with higher growth rates and a higher weight at calving. Hutchison et al. (2017) reported that the younger AFC age groups of 18 to 20 and 22 mo of age have significantly ($P < 0.05$) better calving ease than heifers calving at 24 mo of age, while the groups of heifers calving older than 28 mo of age had generally worse calving ease. Despite the reports in the literature, there was no difference in the calving difficulty between AFC groups ($P = 0.469$) in our meta-analysis.

When calving ease was considered by experiment, the calving ease tended to favor a younger AFC, with 6 experiments showing better calving ease in younger calving heifers compared with only 3 trending toward an increased risk of calving difficulties (Figure 8). The experiment, Hoffman et al. (1996)-2, had the worst calving difficulty outcome for the younger AFC. These heifers were grown at accelerated growth rates but had delayed conception due to not conceiving at first insemination. There is the potential that when heifers have increased growth rates but have a delay in pregnancy, this can result in increased fat deposition, a higher body condition score, and an increase in calving difficulty. In Hoffman et al. (1996), the 2 experimental groups with the worst calving outcomes were the delayed conception groups. These heifers had increased body condition scores at calving when compared with those that conceived on the first in-

semination, with the accelerated growth group having the highest BCS (3.7 out of 5). More experiments are needed to evaluate the effect of AFC on calving difficulty and to evaluate weight at calving and BCS at calving as covariates. From our results, a younger AFC does not increase the risk of calving difficulties.

Longevity is of growing importance to the dairy industry (Dallago et al., 2021). As cattle reach higher lactations, they produce more milk, making them more efficient cattle to feed. Heifer AFC has been investigated as a way of increasing survival and longevity. Our results indicate no difference in survival to calving or the end of first lactation with different AFC. The results are equivocal among experiments. Sherwin et al. (2016), when evaluating survival in 437 UK dairy herds to the end of first lactation, found a significant ($P < 0.05$) increase in risk of herd exit for cattle with an AFC >30 mo. These trends were not evident in our dataset.

One of the main factors for survival to calving and first lactation is the heifer's reproductive performance. Reproductive performance is challenging to compare among studies, as a variety of outcome measures are used to report reproductive performance. There is often a lack of clarity within studies on the exact measures used for each outcome. Attempts have been made to standardize the precision of measures and the nomenclature used (Lean et al., 2016). We have analyzed differing outcomes in subgroups (percentage pregnant and services per pregnancy) and overall to improve the interpretability of the results and allow specific subgroup evaluation. Although reproductive performance of heifers is not a direct outcome of changing AFC, it is important to consider when making management decisions leading to AFC changes. There was no difference ($P = 0.818$) in the reproductive performance of heifers managed to have increased growth or induce different AFC in our study. Breeding heifers to target a younger AFC did not cause additional reproductive loss, suggesting it is a viable management strategy on the farm to lower the AFC. Further evaluation of reproductive performance in other lactations is critical to understand whether there are reproductive benefits or challenges to calving heifers younger.

Lifetime survival could not be evaluated in this study due to the lack of experiments evaluating multiple lactations; only one experiment (Gardner et al., 1988) followed heifers through their lifetime. Other studies reported that as heifers calve younger, there is an increase in the days of herd life without there being necessarily an increase in exit age, leading to increased productive life without necessarily increasing longevity (Curran et al., 2013).

There is a distinct difference between AFC recommendations based on one lactation compared with those based on lifetime production in the literature. When only

first-lactation performance and health were included, recommendations were in the range of 23 to 25 mo (Ettema and Santos, 2004; Nilforooshan and Edriss, 2004), while this is lowered to 21 to 24 mo when lifetime milk yield is considered (Lin et al., 1988; Curran et al., 2013; Hutchison et al., 2017). The differences are due to the cost of decreased production in the first lactation being offset by higher production per day of life when lifetime milk yield is considered, resulting in the recommendation for younger AFC. In our meta-analysis, similar results were observed with a decrease in FLMY and no difference in second and TLMY.

With this model and the available experiments, we cannot evaluate the optimal AFC, as there are limited experiments involving the youngest age groups. Hutchison et al. (2017) found an accelerated decline in FLMY as the AFC went below 21 mo of age, leading to a decreased lifetime milk yield compared with 24 mo of age. The youngest average AFC group evaluated in this study was 618 d (20.6 mo) of age, but no other experiment had an average AFC below 21 mo of age, although 20 experiments involved AFC below the traditional industry standard of 24 mo of age. Increased levels of stillbirths are also an area of concern from experiments with AFC groups below 24 mo of age, with this being a significant financial cost when AFC was evaluated (Ettema and Santos, 2004; Hutchison et al., 2017). This could not be evaluated in this study. The optimal AFC for producers will likely be determined by economic influences such as the cost of increased heifer inputs at younger ages and lost FLMY compared with the gains of reduced rearing time and nonproductive animals on the farm.

The literature on AFC in dairy cattle is lacking prospective randomized trials, especially for modern cattle. The current literature published on AFC mainly involves retrospective enrollment, which could bias results around reproductive performance and production, depending on why the heifers were calved at that age. More prospective experiments investigating the most efficient AFC are needed. It will be important that these experiments include diet, reproductive performance, calving difficulty, weight at calving, growth rates, health outcomes, and production over multiple years to allow for further understanding of this topic.

CONCLUSIONS

Lowering AFC resulted in a small decrease in first-lactation milk, fat, and protein yield, while there was no difference in the heifer reproductive performance, calving difficulty, second- and third-lactation production, and survival outcomes analyzed in this study. More prospective studies are needed to evaluate AFC, especially in younger calving heifers.

NOTES

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Nonstandard abbreviations used: AFC = age at first calving; ES = effect size; FLMY = first-lactation milk yield; PRISMA = Preferred Reporting Items for Systematic Reviews and Meta-Analyses; Q = chi-squared; SLMY = second-lactation milk yield; SMD = standard mean difference; TLMY = third-lactation milk yield.

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